

Predicting Breast Cancer Diagnosis using Supervised Machine Learning



Identifying if the Tumor is Malignant or Benign

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Introduction

Breast cancer is one of the most common cancers among women worldwide. Early diagnosis is critical for successful treatment outcomes. The purpose of this project is to predict whether a breast tumor is malignant (cancerous) or benign(non-cancerous) by using supervised machine learning models.

Dataset

The Breast Cancer Wisconsin (Diagnostic) Data Set originally contained 569 instances and 32 variables: 30 numerical features + 1 target variable (**diagnosis**) + 1 ID variable (**id**) .

There are **30 independent variables**, all on an **(I/R) scale**.

The features are computed from digitized images of fine needle aspirates (FNA) of breast masses. These variables are grouped into three categories:

- **Mean Features** – represent the average value of measurements (e.g., radius_mean, smoothness_mean)
- **SE Features** – represent the standard error of measurements (e.g., radius_se, texture_se)
- **Worst Features** – represent the largest values recorded (e.g., radius_worst, area_worst)

The measured properties of the cell nuclei include:

- **Radius** – distance from center to perimeter
- **Texture** – standard deviation of gray-scale values
- **Perimeter** – length around the boundary
- **Area** – size of the cell nuclei
- **Smoothness** – variation in radius lengths
- **Compactness** – $\text{perimeter}^2 / \text{area} - 1.0$
- **Concavity** – severity of concave portions of the contour
- **Concave points** – number of concave portions
- **Symmetry** – symmetry of the nuclei
- **Fractal dimension** – complexity of the contour

These 10 measurements appear in the three forms, resulting in a total of **30 features**.

Data Preparation

The target column/dependent variable, **diagnosis**, was changed to binary: Malignant = 1, Benign = 0
The **id** column which was removed as it served no predictive purpose. To reduce redundancy, variables with **high correlation** were identified using the Pearson correlation matrix. Any features with a pairwise correlation **greater than 0.95** were removed. The following highly correlated variables were dropped:

- perimeter_mean
- area_mean
- perimeter_worst
- area_worst
- radius_worst
- radius_se

Exploratory Data Analysis

To understand the structure and relationships between different variables in the data, multiple visualization techniques.

The phi-k correlation matrix shows that the features that are most correlated with the diagnosis are, concave points_worst, concave points_mean, radius_mean:

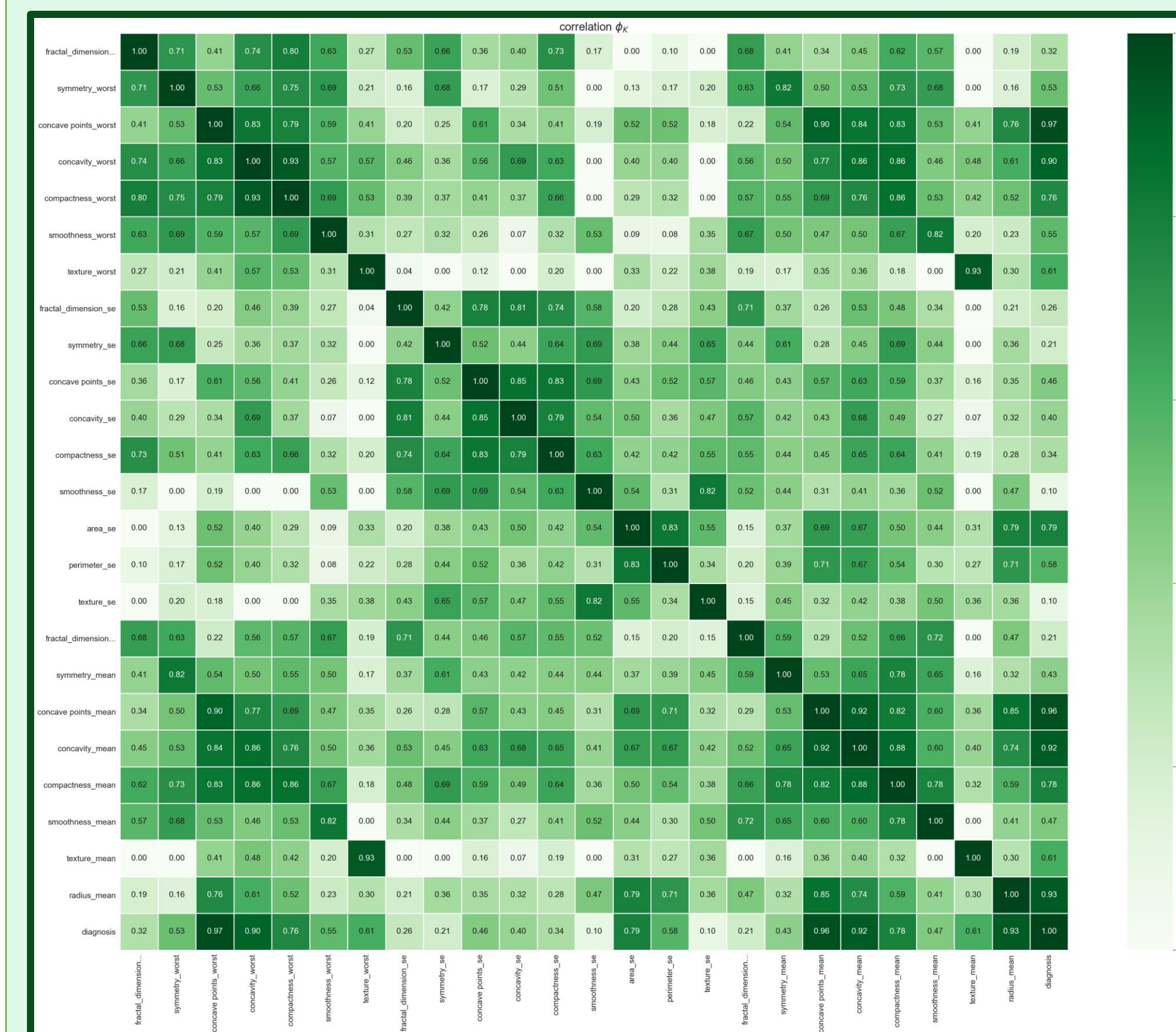


Figure 1: Phi-K Correlation Matrix

Methods

This study purpose was achieved by using multiple machine learning algorithms, and then choosing the most efficient one. The dataset was split into 70% training data and 30% testing data.

Models Used:

- **Decision Tree**
- **Support Vector Machine**
- **Tuned Decision Tree**
- **AdaBoost**
- **Bagging Classifier**
- **Tuned AdaBoost**
- **Tuned Bagging Classifier**
- **Gradient Boosting**
- **Random Forest**
- **Tuned Gradient Boosting**
- **Tuned Random Forest**
- **Support Vector**
- **XGBoost**
- **Logistic Regression**
- **Tuned XGBoost**

They were measured based on the evaluation metrics: accuracy, precision, recall, and F1 score. In this project, **recall was prioritized** over other metrics to ensure that as few malignant cases as possible were missed by the model.

Results

Figure 2: Table showing metric comparisons of all algorithms

Model	Train_Accuracy	Test_Accuracy	Train_Recall	Test_Recall	Train_Precision	Test_Precision	Train_F1	Test_F1
AdaBoost	1.00	0.97	1.00	0.95	1.00	0.97	1.00	0.96
Tuned XGBoost	1.00	0.98	0.99	0.95	1.00	0.98	1.00	0.97
Logistic Regression	0.99	0.95	0.98	0.94	1.00	0.94	0.99	0.94
Tuned AdaBoost	1.00	0.96	1.00	0.94	1.00	0.97	1.00	0.95
Tuned Gradient Boosting	1.00	0.96	1.00	0.94	1.00	0.95	1.00	0.94
XGBoost	1.00	0.97	1.00	0.94	1.00	0.98	1.00	0.96
Bagging Classifier	1.00	0.96	0.99	0.92	1.00	0.97	1.00	0.94
Tuned Bagging Classifier	1.00	0.95	0.99	0.92	1.00	0.95	1.00	0.94
Random Forest	1.00	0.96	1.00	0.92	1.00	0.97	1.00	0.94
Tuned Random Forest	0.97	0.94	0.97	0.92	0.95	0.92	0.96	0.92
Support Vector Machine	0.99	0.96	0.97	0.92	1.00	0.97	0.99	0.94
Gradient Boosting	1.00	0.96	1.00	0.92	1.00	0.97	1.00	0.94
Tuned Decision Tree	0.96	0.92	0.96	0.89	0.93	0.89	0.94	0.89
Decision Tree	1.00	0.92	1.00	0.84	1.00	0.95	1.00	0.89

Most Accurate Model: **Tuned Random Forest**

Figure 3: Tuned Random Forest Confusion Matrix

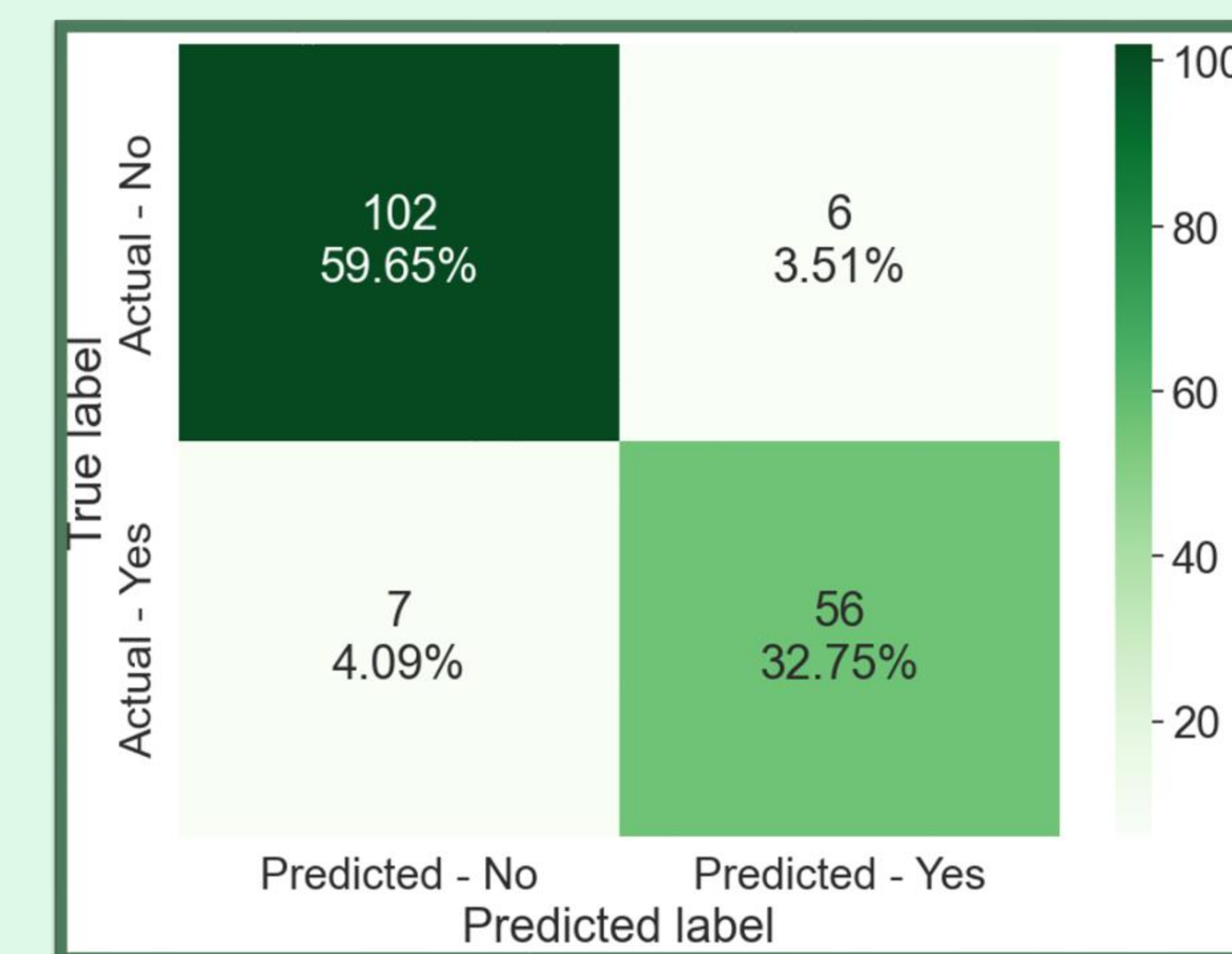
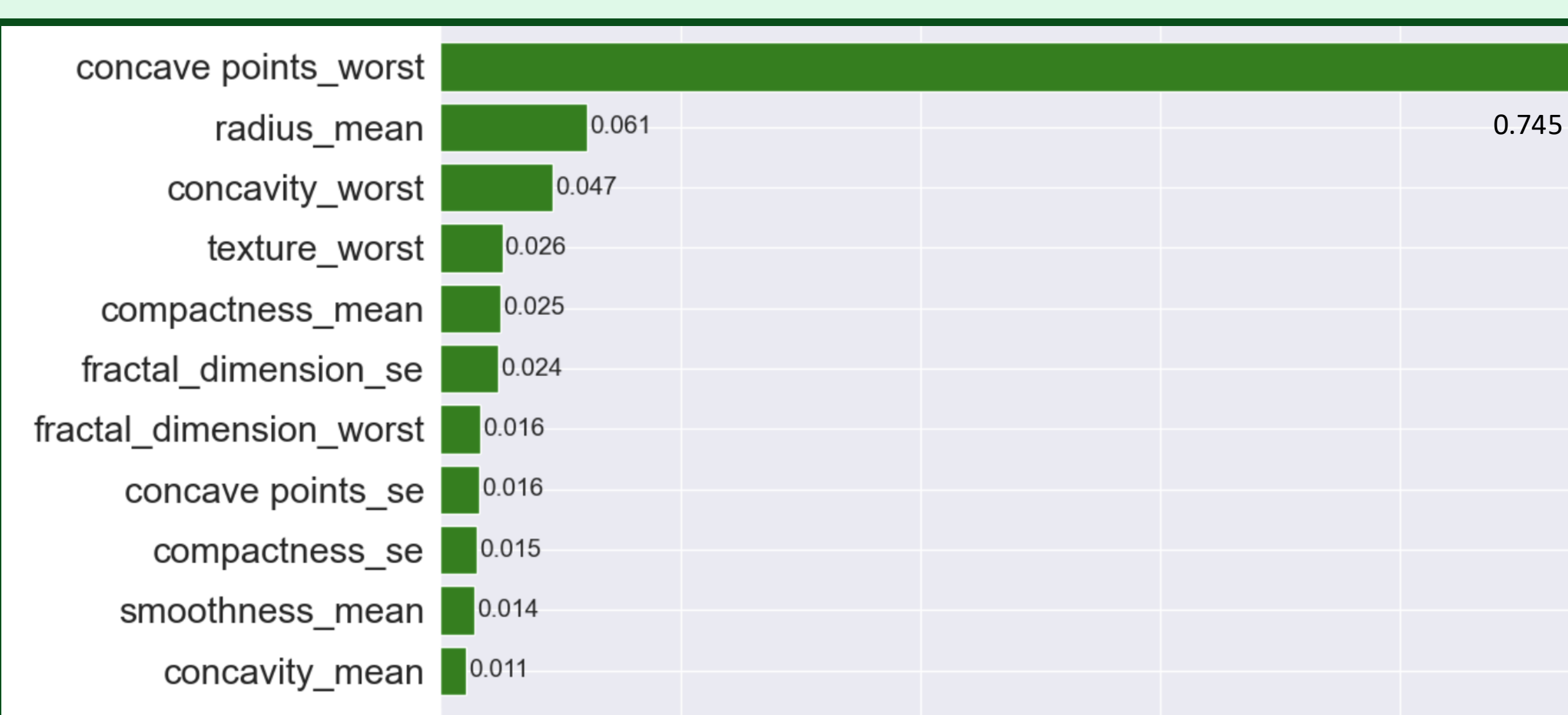


Figure 4: Tuned Random Forest Feature Importance Plot



Conclusion/Insights

- ❑ Best Model: **Tuned Random Forest**
 - ❑ A strong balance between high recall and lower overfitting risk – generalizes to more real-world and unseen datasets.
- ❑ Most Important Evaluation Metric: **Recall**
- ❑ Most Important Features:
 - ❑ **concave points_worst**: Measures the severity and number of concave areas in the tumor outline. Malignant tumors often have irregular, spiky borders.
 - ❑ **radius_mean**: Indicates the average size of the cell nuclei. Malignant tumors typically have larger nuclei with more variability.
 - ❑ **concavity_worst**: Quantifies the depth of concave portions in the tumor shape.
- ❑ Improvement: Many models showed signs of **overfitting**, likely due to the relatively **high number of features (24)** compared to the **sample size (569)**. For a more accurate model, the sample size would need to be increased.

References

1. [Link to Jupyter Notebook](#)
2. Learning, UCI Machine. "Breast Cancer Wisconsin (Diagnostic) Data Set." Kaggle, 25 Sept. 2016, www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data/data.